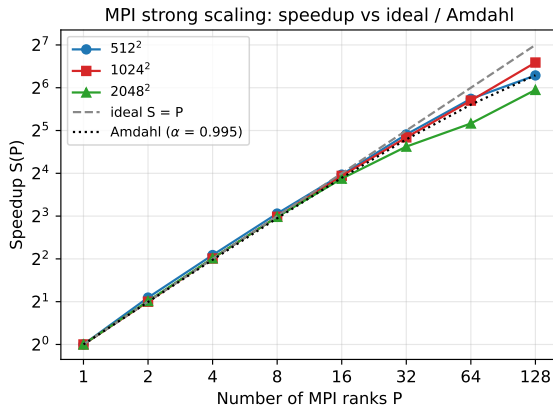


Setup: fixed grid, ranks $P = 1 \dots 128$, three sizes; speedup $S(P) = \text{MLUPS}(P)/\text{MLUPS}(1)$.

- Profiling: serial fraction $s < 1\%$ (only parsing + the single-point probe) $\Rightarrow$ Amdahl predicts *near-linear* speedup.
- We follow it at low P , then bend away at high P .
- The gap is **not** an Amdahl serial fraction (constant) but **communication + memory bandwidth**, which grow in relative weight with P .
- Surface-to-volume: compute $\propto n_y/P$ rows, halo $\propto 2$ rows $\Rightarrow$ thin slabs communicate too much.

Takeaway: deviation from ideal is a hardware effect, not a serial part.



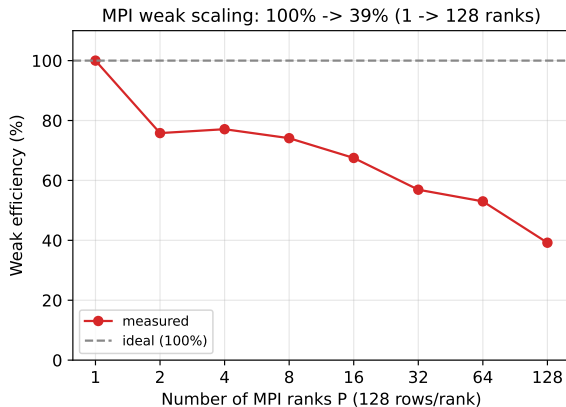
Setup: constant work per rank (128 rows/rank, $n_y = 128P$). Gustafson with tiny s predicts a *flat* efficiency.

- Measured efficiency falls 100% $\rightarrow$ 39%. Three distinct causes:
- (1) $P=1 \rightarrow 2$: a **cache** effect, not comm. The small per-rank domain keeps $P=1$ in the shared L3 (28.6 vs ~ 21 MLUPS for strong $P=1$); at $P=2$ two cores share it and lose the boost.
- (2) $P=8 \dots 64$ (one node): shared **memory-bandwidth saturation** – the dominant limiter, since LBM is memory-bound; *not* the communication.
- (3) $P=128$: crossing onto a **second node** (inter-node network) past 72 cores.

Takeaway: absolute throughput still grows

28.6 $\rightarrow$ 1434 MLUPS.

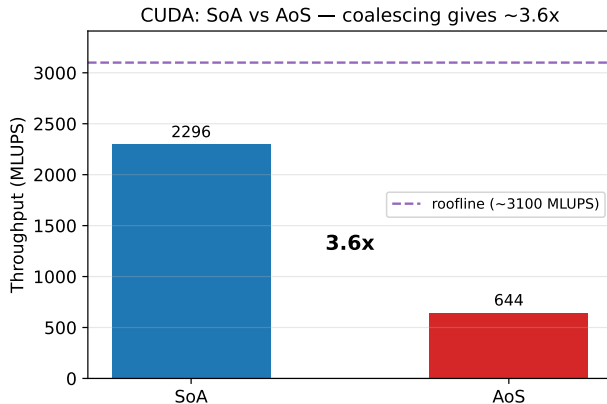
Simon Pierre | MATH-454 | LBM with MPI & CUDA



Design: one thread per cell, one kernel per phase, double buffering for streaming, *no* communication on a single GPU. **2051 MLUPS** $\approx 101\times$ the serial baseline; $St = 0.16$.

- LBM is **memory-bound**: roofline ≈ 3100 MLUPS (288 B/LUP, ~ 900 GB/s). We reach $\sim 66\%$ of it.
- **SoA vs AoS** = $3.6\times$ (the headline): SoA lets a warp *coalesce* its loads; AoS scatters them across cache lines.
- Block-size sweep: broad plateau 128–512 (occupancy, not the ceiling). Size sweep: up to $\sim 81\%$ of the roofline at 4096^2 .

Takeaway: for a memory-bound code the access pattern dominates.



The same strong-scaling data, read as **efficiency** $E(P)$, tells a richer story than “bigger scales better”.

- **Super-linear** at low P ($E > 100\%$): the subdomain starts to fit in **cache**, so each core speeds up – not an error.
- At high P , two *opposite* limiters:
- 512^2 becomes **communication-bound** (4 rows/rank, exchange 2) $\rightarrow 61\%$.
- 2048^2 becomes **bandwidth-bound** (drops $32 \rightarrow 64$ *within* one node) $\rightarrow 56\%$.
- 1024^2 is the **sweet spot** $\rightarrow 75\%$.

Takeaway: trade-off: comm. likes large grids, cache/bandwidth likes small; “bigger always scales better” is only half true (2048^2 is bandwidth-limited).

